

Multiple Choice (10 marks)

1. _____ is the central node of 802.11 wireless operations.
 - (a) WPA
 - (b) Access Point
 - (c) WAP
 - (d) Access Port
2. How does a buffer overflow on the stack facilitate running attacker-injected code?
 - (a) By overwriting the return address to point to the location of that code
 - (b) By writing directly to the instruction pointer register the address of the code
 - (c) By writing directly to %eax the address of the code
 - (d) By changing the name of the running executable, stored on the stack
3. Which of the following styles of fuzzer is more likely to explore paths covering every line of code in the following program?
 - (a) Generational
 - (b) Blackbox
 - (c) Whitebox
 - (d) Mutation-based
4. Local caching of files is common in distributed file systems, but it has the disadvantage that
 - (a) Temporary inconsistencies among views of a file by different machines can result
 - (b) The file system is likely to be corrupted when a computer crashes
 - (c) A much higher amount of network traffic results
 - (d) Caching makes file migration impossible
5. A digital signature needs a
 - (a) Private-key system
 - (b) Shared-key system
 - (c) Public-key system
 - (d) All of them

True / False (10 marks)

Answer True or False

6. True or False: Encryption and decryption do not provide integrity, which ensures data is not altered during transmission.
7. True or False: The addition of the two's-complement numbers 10000001 and 10101010 will result in overflow.
8. True or False: A virus is a hidden part of a program that masquerades as a legitimate application to gain access to a system.
9. True or False: All bit strings with more ones than zeros cannot be described using a regular expression due to their non-regular nature.
10. True or False: The solution to the recurrence $f(2N + 1) = f(2N) = f(N) + \log N$ is $O(N \log N)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the nth digit in the proper fraction of two given numbers.
12. Write a function to group the 1st elements on the basis of 2nd elements in the given tuple list.

End of Paper